Deliverable 1

Questions:

1. What is a web server?

A web server is a **server** set up to host a website or service, receiving and fulfilling requests from **client** machines. Its made up of both hardware and software.

- **Hardware -** The pyhsical machine the server is running on. Were the files and data is stored.
- **Software -** The program or system set up to run the website or service.

2. What are some different web server applications?

1. Apache HTTP Server

An open source web server application. Serves HTTP requests and delivers website content to users. A common hosting service for websites and web applications.

- Operating Systems: Linux, Windows, macOS, Unix-based systems.
- Download/Site: Apache HTTP

2, Microsoft Internet Information Services (IIS)

Developed by Microsoft to allow users to host websites, applications, and web services on Windows and Windows based systems.

- Operating Systems: Windows 10/11
- Project Site: Microsoft IIS
- Download: Microsoft IIS is a built in feature for Windows Operating systems, just requiring enable/disable from IIS menu.

3. Caddy

Another open source web server. Designed to be more streamlined for the user, it stands out by auto-renewing HTTPS certificates.

- Operating Systems: Linux, Windows, macOS, and FreeBSD.
- Download/Site: Caddy

What is Virtualization?

Virtualization is a technology that creates virtual hardware. It simulates physical hardware with software based stand-ins. It allows a single computer to simulate several computers within its hardware.

What is VirtualBox?

VirtualBox is an open-source hypervisor. It is a program to manage multiple virtual machines or VM's.

What is a virtual Machine?

A virtual machine or VM is the simulated computer, managed by a hypervisor, that runs within a physical computer. You can have several Virtual Machines with various hardware settings and environments so long as the physical machine they run on is capable of supporting them.

What does host machine and guest machine mean?

Host Machine:

A host machine is the physical computer or server that will "host" one or more VM's. It provides the physical hardware and resources the guest machine will use.

Guest Machine:

A guest machine are the VM's that run within a physical computer. Guest machines use the host machines resources and can be removed, remade, and tweaked as the user sees fit.

What is Debian?

Debian is an open-source Linux distribution. It is one of the older, more reliable, and common of the Linux Distro's that can be found. It is notable for being one of the more trusted, and stable distro's out there.

What is a firewall?

A firewall is a network security system. Its sort of like a security checkpoint between an internal network and the unknown and unverified external network. A firewall scans and monitors incoming traffic for malicious, suspicious, or unauthorized activity.

What is SSH?

SSH is Secure Shell. It is a protocol for sharing access and data between two computers or servers. It uses encryption keys to authenticate and verify identity between the two nodes, allowing the secure connection even via unsecure networks such as the internet.

What is an IP Address?

An IP Address, or Internet Protocol Address, is a collection of digits that represent the address and ID of a single computer or device. The number of digits in an IP Address varies depending on which protocol you are using:

- IPv4: **Example - **123.123.1.123 - 12 digits. The more common and older version of IP.
- IPv6: The newer and more robust version of IP. It uses 8 groups of hexadecimal numbers to provide a **much** broader number of address's to utilize for individual devices.

What is a network mask?

A network mask is used in conjunction with an IP Address to help identify which network a device belongs too.

What is a port?

A port is an access point on a computer that can allow another device to send or receive data. Ports are designated for specific purposes or applications.

What is port forwarding?

Port forwarding is a method of directing incoming network traffic after it accesses a port. It is how a request from a client machine is directed to the specific requested device or service.

What is localhost?

Localhost is the name given to represent the current computer or device being used. It allows the computer to communicate with itself and is often used to test applications, websites, and services before pushing them live to the wider internet.

What does the IP Address `127.0.0.1` represent?

The IP address `127.0.0.1` represents localhost. It is an IP address that only accesses something locally on your computer. This is how developers test the above mentioned websites, applications, and services by creating a local server that can only be accessed by the host computer.

What is Git?

Git is a type of version control software that helps users track, record, and manage changes to a project, file, or system locally. Not to be confused with github.

What is Github?

Github is the cloud management version of Git. It interfaces with a computers local Git repositories, and stores it remotely on Github's server. This allows larger teams to work on different parts simultaneously while the project is hosted on Github. Github provides collaborative tools such as push or pull requests, which essentially allows users to upload and propose changes or modifications for review by users with appropriate permissions.